

FRONT-END DEVELOPMENT

ASSIGNMENT 3 TEMPLATE

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YEAR: II

COLLEGE CODE:1455

COLLEGE NAME: Tagore College of Arts and Science

DEPARTMENT: Computer Science

SEMESTER: IV

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DATE: 10/03/2026

PROJECT TITLE: JavaScript Interactivity for Event Registration

1. Problem Statement:

Small event organizers need a way to ensure the data they collect is accurate and to see registrations update instantly without refreshing the page. While Assignment-2 focused on look and feel, this assignment adds functionality to validate user input and dynamically manage participant lists.

2. Objective:

To implement JavaScript for client-side validation and DOM manipulation. The goal is to ensure only correct data is submitted and to provide immediate visual feedback by updating the "Registered Students" table automatically.

3. Requirements:

This section details the functional requirements implemented to move the project from a static design to an interactive web application:

1. Form Validation:

- **Mandatory Checks:** Wrote conditional logic in script.js to verify that Name, Date of Birth, and College fields are not empty.
 - **Format Verification:** Used string methods to check for the presence of the "@" symbol in the Email field and ensured the Phone number contains exactly 10 digits.
 - **Selection Logic:** Added checks to confirm that an event has been chosen from the dropdown and that both Gender and Mode radio buttons have a "checked" status.
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3.2 Confirmation Checkbox Validation:

- **Prerequisite Check:** Implemented an if statement to check the .checked property of the declaration checkbox.
 - **Error Prevention:** If the checkbox is unselected, the script triggers a warning alert and halts the submission process to ensure user agreement.
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3.3.Dynamic Display of Registered Participants:

- **Data Extraction:** Programmed the script to capture values from textboxes, selected radio buttons, and the dropdown menu simultaneously upon submission.
 - **DOM Table Updates:** Used the insertRow() and insertCell() methods to create a new row at the bottom of the existing table for every successful registration.
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3.4 Prevent Page Reload:

- **Event Interception:** Utilized the e.preventDefault() method within the form's onsubmit function.
 - **Seamless Experience:** This prevents the browser from refreshing or navigating away, allowing the participant table to update instantly while keeping all current session data intact.
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3.5 Clear Form After Successful Submission:

- **State Reset:** Used the .reset() method on the form object to clear all input fields after the data has been safely moved to the table.
 - **Validation Cleanup:** Added a loop to remove "redBorder" or "greenBorder" CSS classes from all inputs, returning the form to its original blank state.
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3.6 Display Success Message:

- **Visual Confirmation:** Directed a "Registration Successful!" message to a dedicated statusMsg div with green text styling.
 - **Auto-Removal:** Applied setTimeout() to automatically clear the success message after 3 seconds to keep the interface clean.
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3.7 Input Field Feedback:

- **Visual Cues:** Programmed the script to dynamically assign CSS classes (redBorder for errors or greenBorder for valid data) based on validation results.
 - **Instant Correction:** This provides the user with immediate visual confirmation of which fields need correction before they can proceed.
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3.8 Optional Enhancements:

- **File Upload Validation:** Included a check to ensure an ID proof file has been selected before the form is considered complete.
- **Consistent Formatting:** Ensured that the dynamically added data in the table maintains the same professional alignment and styling as the header row.

4. JavaScript Concepts Used:

- **DOM Selection:** Used document.getElementById to access form inputs and the table.
- **Event Listeners:** Applied .addEventListener('submit', ...) to handle user actions.
- **Conditional Logic:** Used if-else statements to check for valid input formats.
- **Array/Object Handling:** Captured form values into variables before inserting them into the table.
- **Function Definitions:** Created reusable functions for validation and table updates.

5. Implementation Steps:

Script Link: Connected the external script.js to the HTML file.

Validation Logic: Wrote functions to verify email patterns and phone number lengths.

Submission Handler: Prevented the form's default reload behavior to keep the dynamic data visible.

Table Insertion: Used insertRow() and insertCell() to add the new participant details to the bottom table.

Cleanup: Triggered the reset() method to prepare the form for the next user

6. Output:

InterCollege Technical Workshop

Register for our upcoming technical Workshop.

Date: 15-03-2026

Time: 10:00 AM - 1:00 PM

Venue: Seminar Hall

Organizer: Department of Computer Science

Personal Info

Name: Hariharan s

Email: test

Phone: 8056194886

DOB: 10/26/2005

Gender: ☒ Male ☐ Female

Event Details

Event: Java Workshop

Mode: ☐ Online ☒ Offline

Additional Details

College: Tagore College od Arts and

Upload ID: collegeID.jpeg

Comments(optional):

☐ I confirm all details are correct.

This page says

Check all fields and the checkbox!

Upload ID: collegeID.jpeg

Comments(optional):

☐ I confirm all details are correct.

Registered Students

Name	Email	Phone	Event	Mode
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Contact details:

Name: Hariharan S

Email: kumaravelat2005@gmail.com | Phone: 8056194886

Help desk timing : 8.30 - 2.30 (only on working daya)

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InterCollege Technical Workshop

Register for our upcoming technical Workshop.

Date: 15-03-2026	Time: 10:00 AM - 1:00 PM
Venue: Seminar Hall	Organizer: Department of Computer Science

Personal Info

Name:

Email:

Phone:

DOB:

Gender: ☒ Male ☐ Female

Event Details

Event:

Mode: ☐ Online ☒ Offline

Additional Details

College:

Upload ID: collegeID.jpeg

Comments(optional):

☒ I confirm all details are correct.

Before submit

After submit

Event Details

Event:

Mode: ☐ Online ☐ Offline

Additional Details

College:

Upload ID: No file chosen

Comments(optional):

☐ I confirm all details are correct.

Registration Successful!

Name	Email	Phone	Event	Mode
Hariharan s	kumaravelat2005@gmail.com	8056194886	Java	Offline

Contact details:

Name: Hariharan S

Email: kumaravelat2005@gmail.com | Phone: 8056194886

Help desk timing : 8:30 - 2:30 (only on working daya)

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7.Learning Outcome:

Interactivity: I learned how to move beyond static styling and create a functional web application using JavaScript.

Data Integrity: I understood how to enforce rules on user input to ensure the database (or table) only receives correctly formatted information.

DOM Skills: I gained confidence in manipulating the Document Object Model to change page content dynamically without a server.

8. Conclusion:

The project successfully integrates HTML, CSS, and JavaScript to form a complete registration system. The validation logic ensures data quality, while the dynamic table update provides immediate satisfaction to the user, proving that client-side scripting is essential for a modern web experience.